

FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

TLDR

Two hours finding out what actuators feel like, and leaving able to drive one from your own code.

Get a board and run task 01. Something spins.

Change two numbers, upload again. That loop is the workshop.

Pick up the actuators you are not driving.

Work through as many of the five tasks as you get to.

Build one signal someone else can read without looking.

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the tables at the front

- Go and feel the ones your task does not use

Finish by building one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Boards out, task 01 running for everyone
0:25	60 min	Tasks 02 to 05, at your own pace
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down

FIVE TASKS

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

- 01 Make something spin coin motor**
- 02 Make something tap solenoid**
- 03 Make something warm Peltier tile**
- 04 Make it notice you a wire and foil**
- 05 Make it answer back both together**
- 06 Use what you have your own phone**

1 to 3 are output. 4 is input. 5 is both. 6 needs no board.

Nobody is expected to finish all five.



HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one



github.com/gust1527/multimodal-actuator-workshop

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DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT



WHAT YOU GET IN THE BASE KIT

One set per pair. Collect it before you start task 01.

Board, breadboard, jumper wires, USB cable

- Assorted resistors and a few transistors

Breadboard power supply, drawer 8A

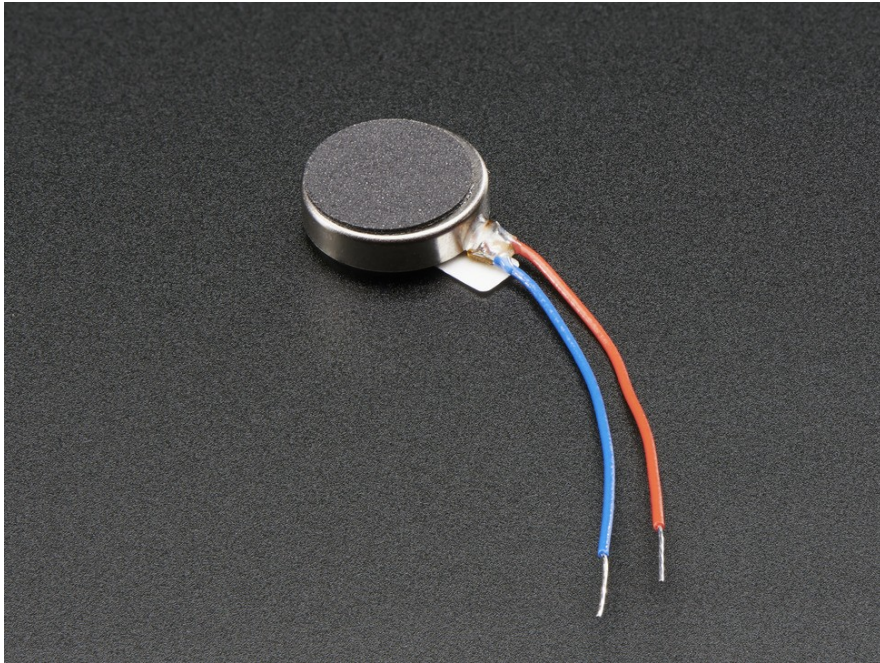
- 42 in stock. Use it for anything that is not just an LED.

L9110S H-bridge, drawer 2C

- 24 in stock. Needed for the Peltier and for reversing a motor.

WHAT IS ON THE TABLES

VIBRATING MINI MOTOR DISC



An off-centre weight on a motor shaft, sealed in a disc. The same part as in your phone.

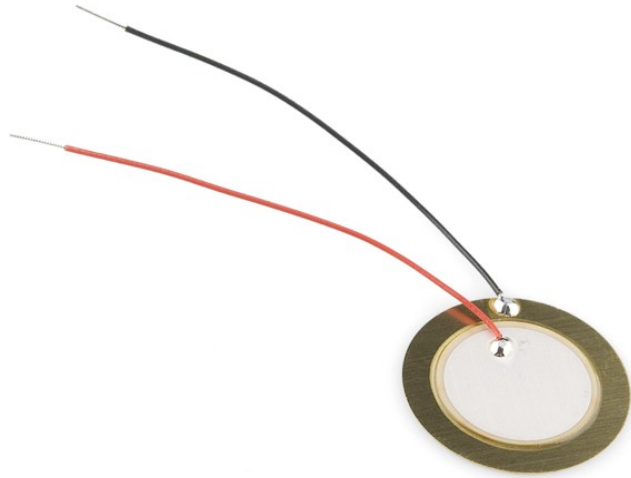
Feels like: a buzz you cannot make gentle and fast at once.

USED FOR

Phone notifications, controller rumble, anything worn under clothing.

Drawer 1E | 117 in stock | 2.5-3.8 V | PWM pin

PIEZO ELEMENT



A crystal that flexes when you put current across it. Near-instant, almost no power, but shallow.

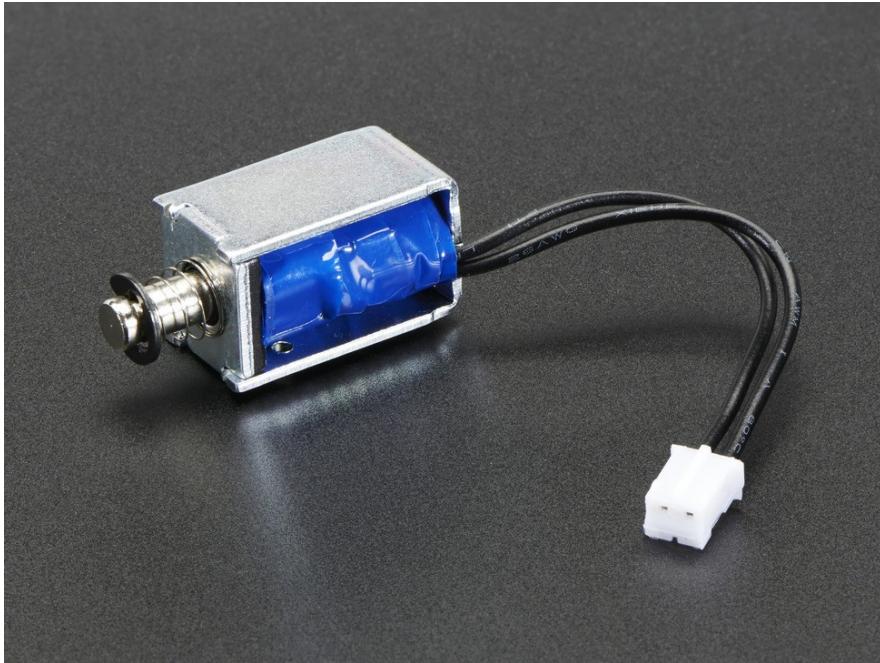
Feels like: a tick, not a thump.

USED FOR

Clicks and confirmations. Anything where lag would feel broken.

Drawer 6F | 69 in stock | any digital pin

MINI PUSH-PULL SOLENOID



A coil that yanks a metal rod when you energise it. No half a tap: it fires or it does not.

Feels like: a person tapping you, not a machine signalling.

USED FOR

Navigation cues on the body, braille cells, alerts in noise.

Drawer 3E | 24 in stock | 5V | needs a MOSFET

CAPACITIVE PAD



A wire taped behind foil, card or fabric. No sensor, no breakout board.

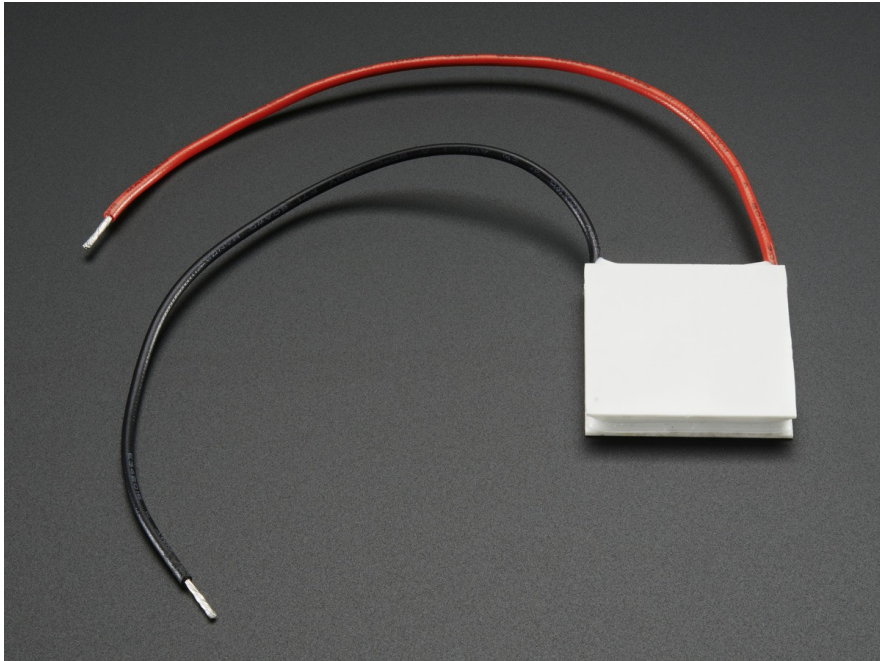
Feels like: nothing. That is the point. The surface stays plain.

USED FOR

Invisible controls in wood or fabric, waterproof panels.

No part number. A wire, on any analogue pin

PELTIER TILE



Heats one face and cools the other. Flip the current and it reverses.

Feels like: slow. Several seconds before you are sure which way.

USED FOR

Slow ambient state. Never an alert.

Drawer 4C | 25 small, 35 big | needs an H bridge

SURFACE TRANSDUCER



Turns any surface into a speaker. Drive it so one frequency is felt and another is heard, from the same part.

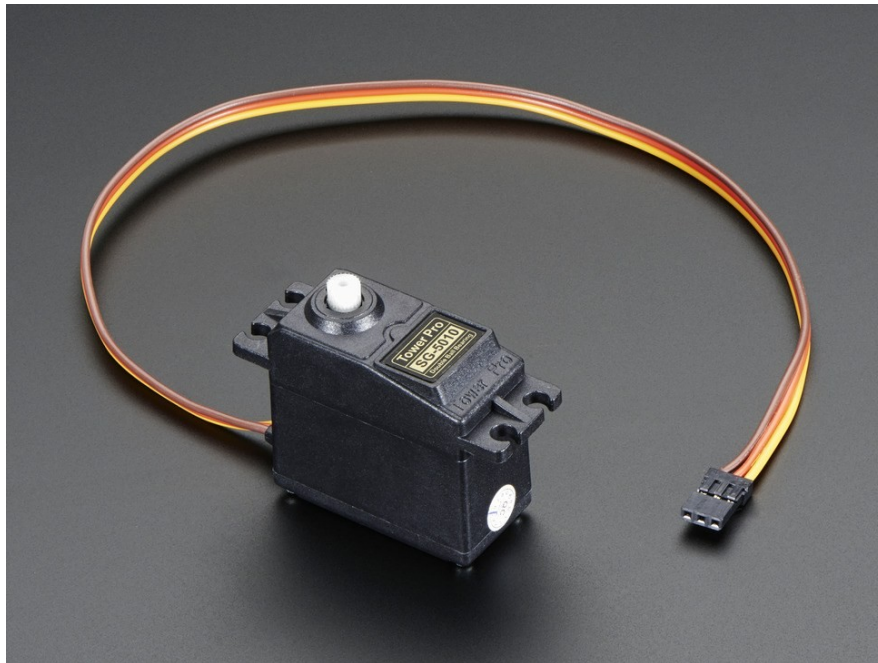
Feels like: more certain, rather than louder.

USED FOR

Noisy or bright environments, and accessibility.

Drawer 7F | 12 large, 13 medium, 23 bone | needs an amp

SERVO



A motor that holds a position instead of spinning freely. Tell it an angle and it goes there.

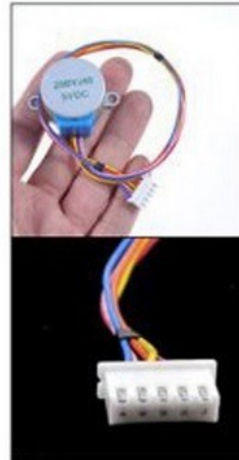
Feels like: something deliberate moving, with force behind it.

USED FOR

Anything that points, pushes, opens or resists a hand.

Drawer 2F | 80 in stock | Servo.h | needs its own 5V

STEPPER MOTOR



Moves in fixed steps rather than continuously, so you always know where it is without a sensor.

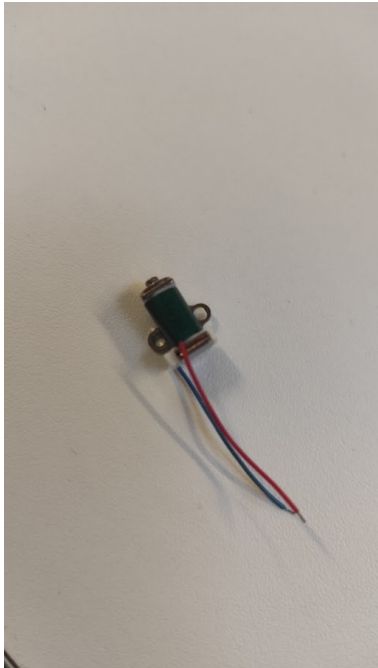
Feels like: precise, and audibly clicky.

USED FOR

Slow accurate motion. Dials, sliders, anything positioned.

Drawer 2A | 44 in stock | needs a ULN2003 driver

ELECTROMAGNET



Grabs and releases ferrous metal on command. No moving parts of its own.

Feels like: a grip that appears and vanishes.

USED FOR

Latches, holds, and anything that should let go on cue.

Drawer 3D | 17 mini, 11 standard | needs a MOSFET

FREESTYLE IS ENCOURAGED

You do not have to do the tasks in order.

You do not have to finish them.

Getting task 01 working and then properly playing with task 02 beats rushing through all five.

If you want to drive something that is not in front of you, go and get it.

BEFORE YOU POWER ANYTHING

The solenoid and the Peltier both pull real current.

- They need their own supply. Ask before you rewire either.

The Peltier needs its heatsink on. Every time.

- Without it the tile cooks itself in about a minute.

Never drive a motor straight from a pin.

- 40 mA limit. The motor wants nearly twice that.

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

THE SENSORS ALREADY IN THE ROOM

Every phone here is a multimodal device. Some of it the browser will hand you, some of it it will not.

Works in a web page today

- Accelerometer and gyroscope. Tap to grant, HTTPS only
- Microphone. Easiest continuous input there is
- Camera. Even reads a pulse off a face, badly
- Trackpad pressure. Pointer Events carry a 0 to 1 force

Does NOT work, whatever the tutorial says

- Phone vibration on iOS. WebKit has never shipped it
- Apple Watch or Garmin, live. Both need a native app

THE ONE TO CHECK BEFORE YOU PROMISE IT

`navigator.vibrate`: Android yes, iOS no, at every version to date.

`caniuse` lists Safari 3.1 to 27 as unsupported.

Chrome and Firefox on iOS are Safari underneath, so they inherit the gap.

"The phone buzzes" therefore works on some of the room and no iPhones.

Check the feature, not the tutorial, before it is in your plan.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

Blind test: they name all three



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- Note which two got confused, and why

MULTIMODAL INTERACTION
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THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.

TEST IT

Hand your device to another group with the screen turned away.

- They can tell all three messages apart.
- It does not fire continuously when held.
- You know which two got confused, and what would have separated them.
- It still works when they are not looking at it.

WHAT YOU ACTUALLY LEARN

Three things you cannot get from a datasheet.

What things feel like

- A motor is a nudge. A solenoid is a knock. A Peltier is slow.

How to drive one

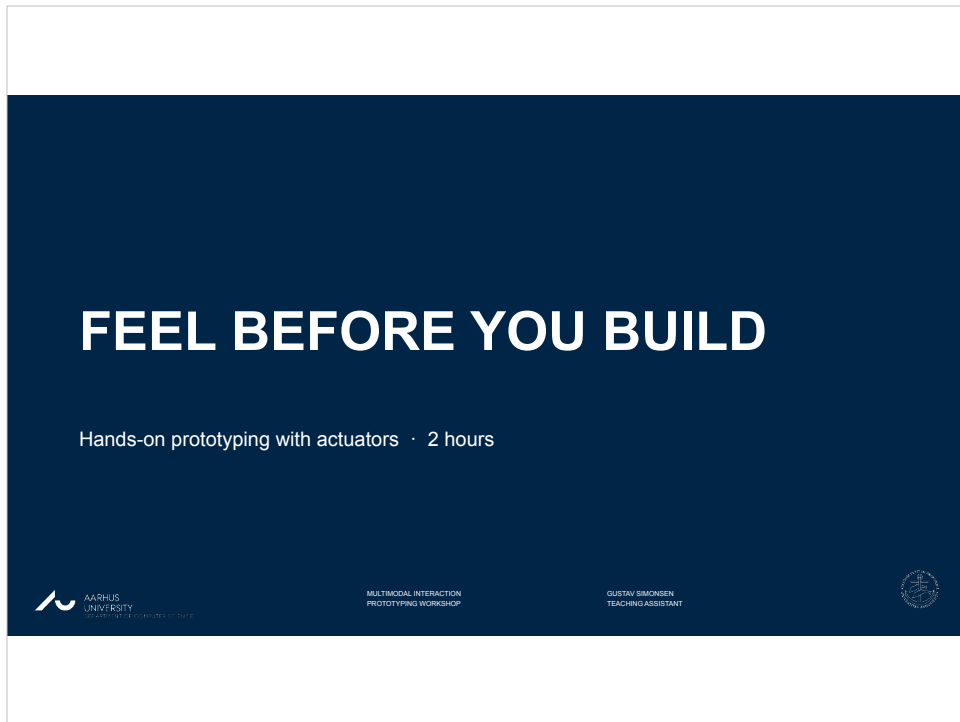
- PWM, direction, debouncing, and why a motor never goes straight to a pin.

What to pick

- Choosing the actuator before designing the interaction, not in week 46.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop



0:00. Start here, and do not read the slide.

Open with the demo, not with talking. Have the coin motor and the solenoid both wired and running before anyone sits down.

Buzz a phone-style notification on the motor. Then send the same message on the solenoid. Ask which felt more urgent. Let them answer. Nobody needs theory for that, and it is the argument for the whole session.

TLDR

Two hours finding out what actuators feel like, and leaving able to drive one from your own code.

Get a board and run task 01. Something spins.

Change two numbers, upload again. That loop is the workshop.

Pick up the actuators you are not driving.

Work through as many of the five tasks as you get to.

Build one signal someone else can read without looking.



Fifteen seconds. It is a map, not a talk.

Useful because half the room arrives expecting a lecture and needs to be told immediately that this is not one.

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46



Say the full version out loud, the slide only carries the skeleton:

"You can read a datasheet. You have never held one of these. Today you put a vibration motor against your own wrist, decide it feels cheap, and pick something else. Better now than in week 46 with the demo three days out."

Every year projects end up on the screen and the speaker, and touch gets designed on paper then dropped. Two minutes, then move.

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the tables at the front

- Go and feel the ones your task does not use

Finish by building one signal

- Someone else has to read it



Set the shape clearly, because it is not a lecture and they will wait to be lectured at.

A board each. Five tasks. Everything pre-wired. They upload, change numbers, upload again.

The actuators they are NOT driving sit on tables around the room. Tell them explicitly to go and pick those up between tasks. The comparison still happens, just on their own initiative rather than on a timer.

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Boards out, task 01 running for everyone
0:25	60 min	Tasks 02 to 05, at your own pace
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down



Show it, do not narrate it. Fifteen seconds.

The one hard checkpoint is 0:25. Everyone should have task 01 running by then.

If someone is still fighting drivers at 0:30, pair them with a group that is working rather than debugging it alone.



0:10. Boards out.

Get EVERYONE through task 01 before anyone races ahead. It is the shortest and it flushes out the setup problems: wrong board, wrong port, missing driver.

Once task 01 runs for a pair, they can move at their own pace.

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

- | | | |
|----|---------------------|-----------------|
| 01 | Make something spin | coin motor |
| 02 | Make something tap | solenoid |
| 03 | Make something warm | Peltier tile |
| 04 | Make it notice you | a wire and foil |
| 05 | Make it answer back | both together |
| 06 | Use what you have | your own phone |

1 to 3 are output. 4 is input. 5 is both. 6 needs no board.



Nobody is expected to finish all five.

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PROTOTYPING WORKSHOP

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TEACHING ASSISTANT



Ninety seconds on this slide, no more.

Name the arc, it is the point: 1 to 3 are output, 4 is input, 5 is both. By task 5 they have built a complete interaction loop, which is the shape of most project work.

Say clearly that nobody is expected to finish all five. Getting 01 working and then properly playing with 02 beats rushing all of them.

Task 05's last experiment IS the closing brief, so the work carries over.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one



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Say the format once here rather than repeating it per task.

CHANGE ME at the top: two or three numbers, already set to something that works. THINGS TO TRY at the bottom: four experiments, easy to hard, and the last one usually asks for a rewrite rather than a tweak.

Upload problems, in the order they will hit you:

- * wrong board or port under Tools
- * ADCTouch not installed (tasks 04 and 05)
- * solenoid task: board resets when it fires means its supply is off

WHAT YOU GET IN THE BASE KIT

One set per pair. Collect it before you start task 01.

Board, breadboard, jumper wires, USB cable

- Assorted resistors and a few transistors

Breadboard power supply, drawer 8A

- 42 in stock. Use it for anything that is not just an LED.

L9110S H-bridge, drawer 2C

- 24 in stock. Needed for the Peltier and for reversing a motor.



PROTOTYPING WORKSHOP

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Hand this out before anything else, or you will spend the session fetching parts one at a time.

One set per pair. The three drawer numbers are the ones people actually come back and ask for: 8A breadboard supply, 2C H-bridge, 11F amplifier.

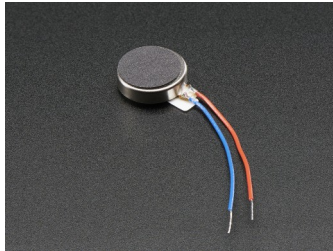
Jumper wires, breadboards and resistors are not tracked in the component database, so make sure there are enough on the tables before students arrive.



Two minutes total for this whole run of slides. It is a menu, not a lecture: they are choosing what to go and pick up later.

Hold up the real part as you show each one. The photo is a reminder for people reading the deck afterwards; the object in your hand is what lands in the room.

VIBRATING MINI MOTOR DISC



An off-centre weight on a motor shaft, sealed in a disc. The same part as in your phone.

Feels like: a buzz you cannot make gentle and fast at once.

USED FOR

Phone notifications, controller rumble, anything worn under clothing.

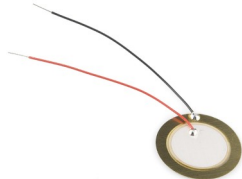
Drawer 1E | 117 in stock | 2.5-3.8 V | PWM pin



The one everybody already owns without knowing it. Every phone has one.

The point worth making: speed and strength are welded together. That single constraint is why the other five exist.

PIEZO ELEMENT



A crystal that flexes when you put current across it. Near-instant, almost no power, but shallow.

Feels like: a tick, not a thump.

USED FOR

Clicks and confirmations. Anything where lag would feel broken.

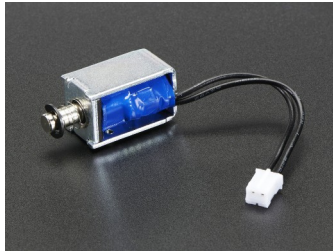
Drawer 6F | 69 in stock | any digital pin



The escape from the coin motor's compromise.
A crystal that flexes when
you put current across it: near-instant, almost
no power, but shallow.

A tick, not a thump. Point anyone here who
needs something to feel responsive
rather than just present. 69 in the drawer, so
nobody has to share.

MINI PUSH-PULL SOLENOID



A coil that yanks a metal rod when you energise it. No half a tap: it fires or it does not.

Feels like: a person tapping you, not a machine signalling.

USED FOR

Navigation cues on the body, braille cells, alerts in noise.

Drawer 3E | 24 in stock | 5V | needs a MOSFET



Pass this one around if you can. It is the one people remember, because a knock reads as a person rather than a machine.

Warn them now that it needs its own supply, so it does not come as a surprise at task 02.

CAPACITIVE PAD



A wire taped behind foil, card or fabric. No sensor, no breakout board.

Feels like: nothing. That is the point. The surface stays plain.

USED FOR

Invisible controls in wood or fabric, waterproof panels.

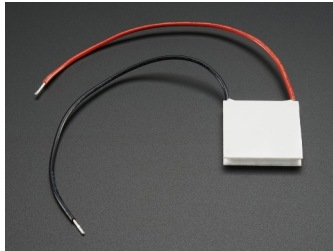
No part number. A wire, on any analogue pin



Hold up a wire and a piece of foil. That is genuinely the whole sensor.

This is the slide that changes what people think is possible, because it costs nothing and hides completely inside a prototype.

PELTIER TILE



Heats one face and cools the other. Flip the current and it reverses.

Feels like: slow. Several seconds before you are sure which way.

USED FOR

Slow ambient state. Never an alert.

Drawer 4C | 25 small, 35 big | needs an H bridge



Say the honest limitation out loud: it is slow,
and hot alternating with
cold just reads as lukewarm.

Good for slow ambient state. Wrong for alerts.
Saying so here saves a project
from finding out in week 46.

SURFACE TRANSDUCER



Turns any surface into a speaker. Drive it so one frequency is felt and another is heard, from the same part.

Feels like: more certain, rather than louder.

USED FOR

Noisy or bright environments, and accessibility.

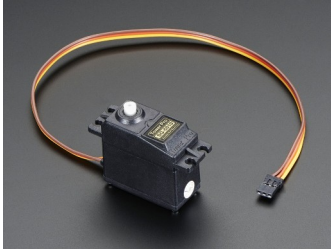
Drawer 7F | 12 large, 13 medium, 23 bone | needs an amp



The redundancy argument. Same driver, one frequency you feel and one you hear.

Together is not louder, it is more certain.
Cheapest reliability a project can buy, and it is also the accessibility answer.

SERVO



A motor that holds a position instead of spinning freely. Tell it an angle and it goes there.

Feels like: something deliberate moving, with force behind it.

USED FOR

Anything that points, pushes, opens or resists a hand.

Drawer 2F | 80 in stock | Servo.h | needs its own 5V



Worth showing even though it is not in a task.
Most projects that need
something to physically move end up here
rather than on a vibration motor.

80 in stock, so nobody has to share.

STEPPER MOTOR



Moves in fixed steps rather than continuously, so you always know where it is without a sensor.

Feels like: precise, and audibly clicky.

USED FOR

Slow accurate motion. Dials, sliders, anything positioned.

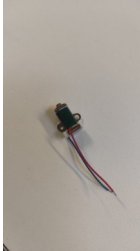
Drawer 2A | 44 in stock | needs a ULN2003 driver



The precision option. Moves in known steps, so you can position it without any feedback sensor at all.

Needs a ULN2003 driver board, drawer 2A, 27 of them.

ELECTROMAGNET



Grabs and releases ferrous metal on command. No moving parts of its own.

Feels like: a grip that appears and vanishes.

USED FOR

Latches, holds, and anything that should let go on cue.

Drawer 3D | 17 mini, 11 standard | needs a MOSFET



The one people forget exists. It grabs and lets go, with nothing moving.

Good answer for anything that should hold and then release: a latch, a door, a thing that falls when the event happens.

FREESTYLE IS ENCOURAGED

You do not have to do the tasks in order.

You do not have to finish them.

Getting task 01 working and then properly playing with task 02 beats rushing through all five.

If you want to drive something that is not in front of you, go and get it.



Say this out loud, do not just show it. Some students will not believe they are allowed to skip ahead or stop early unless you tell them.

The failure mode is a pair rushing all five tasks badly instead of doing two properly. Name that explicitly.

BEFORE YOU POWER ANYTHING

The solenoid and the Peltier both pull real current.

- They need their own supply. Ask before you rewire either.

The Peltier needs its heatsink on. Every time.

- Without it the tile cooks itself in about a minute.

Never drive a motor straight from a pin.

- 40 mA limit. The motor wants nearly twice that.



The only slide where you should sound firm.

Three real hazards:

- * solenoid and Peltier on their own supply, not USB
- * Peltier heatsink on before it is switched on, no exceptions
- * no motor straight from a pin

Say the numbers: a pin gives 40 mA, the motor wants about 75. It is not a rule they have to take on faith.

Also mention that a board resetting when the solenoid fires is a POWER problem, not a code bug. It will save someone twenty

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

`reading > baselineAverage() * 1.01`

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

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PROTODIGITAL INTERACTION
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Slow down here. This is the single idea most worth stealing from today, and it is worth interrupting the room for once most people have reached task 04.

Two things students get wrong:

1. A fixed threshold. Works in the morning, fails after lunch, because the reading drifts with humidity and mains hum.

2. Feeding the baseline buffer always. A long press then teaches the baseline that a finger is normal, and the touch silently stops working. The buffer must only learn while untouched

INPUTS, AND PRIOR ART



1:25. Pull the room back together.

One sentence per group: what surprised you.
No slides, no laptops. Ten minutes.

Expect someone to report the Peltier delay. Let
that land, it is far more
persuasive from a peer than from you.

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all



Mention, do not teach. These sit on one table with a sign.

The honest warnings matter more than the specs: PPG is useless once the hand moves, GSR drifts so only relative change works, FSRs need per-pad calibration.

Push the phone option hard. For a lot of projects it beats everything else on the table and needs no soldering at all.

THE SENSORS ALREADY IN THE ROOM

Every phone here is a multimodal device. Some of it the browser will hand you, some of it it will not.

Works in a web page today

- Accelerometer and gyroscope. Tap to grant, HTTPS only
- Microphone. Easiest continuous input there is
- Camera. Even reads a pulse off a face, badly
- Trackpad pressure. Pointer Events carry a 0 to 1 force

Does NOT work, whatever the tutorial says

- Phone vibration on iOS. WebKit has never shipped it



• Apple Watch or Garmin, live. Both need a native app

PROTOTYPING WORKSHOP

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Worth five minutes, because it reframes what a prototype can be. Half the room assumes multimodal means buying hardware.

Have them open the page on their own phones while you talk. The split between what works and what does not lands much harder when it is their device.

The watch line matters for project planning: several groups every year propose live Apple Watch or Garmin data, and neither streams to a web page. Both need a native companion app. Fine for after-the-fact analysis, not for anything

THE ONE TO CHECK BEFORE YOU PROMISE IT

navigator.vibrate: Android yes, iOS no, at every version to date.

caniuse lists Safari 3.1 to 27 as unsupported.

Chrome and Firefox on iOS are Safari underneath, so they inherit the gap.

"The phone buzzes" therefore works on some of the room and no iPhones.

Check the feature, not the tutorial, before it is in your plan.



The single most useful fact on the slide, and the one people get wrong because a blog post said otherwise.

Say it plainly: iPhones do not vibrate from a web page. Not with a polyfill, not with a library, not in Chrome for iOS, which is Safari underneath.

The general lesson is bigger than vibration: check the feature on caniuse before it goes in a project plan, not after. Two minutes now against a rewrite in week 46.

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

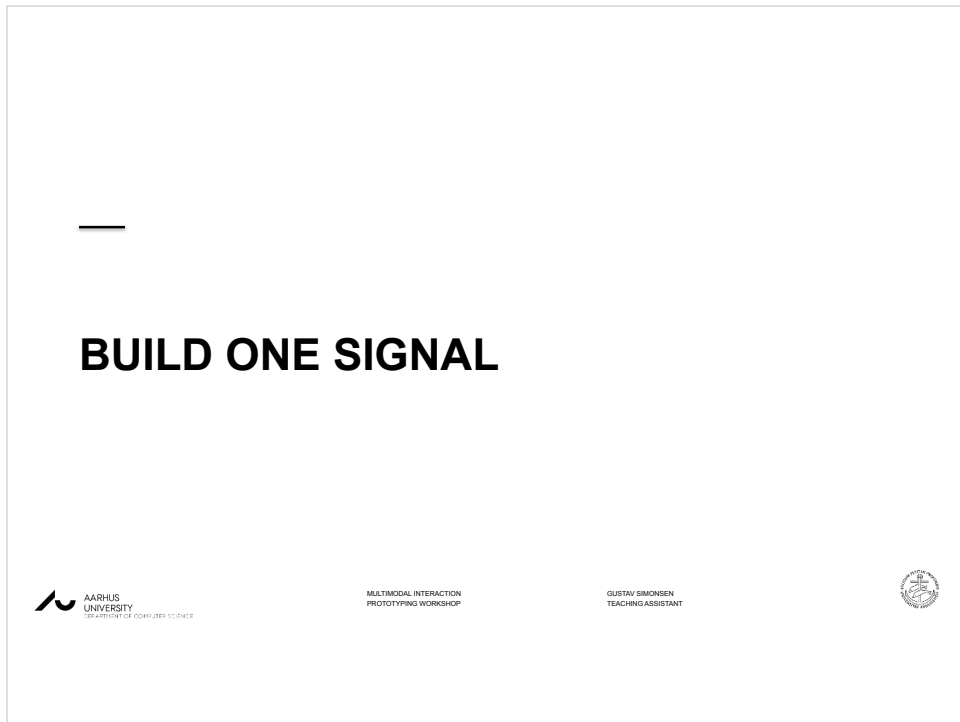


Five minutes, and put the actual repos on screen if you can.

The argument: this was built by students at your stage, not by a lab with a budget. The gap between "we should use touch" and "it moves when you touch it" is smaller than it looks.

The moving screen has a browser control panel over the serial port, which is what you will wish you had at the project demonstrations.

Note for you: the sock repo has a hardcoded Wi-Fi password and live patient IDs



1:35. The blind test.

Most groups will already have something from task 05. This is where it gets tested by someone who did not build it.

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are



Blind test: they name all three

- Note which two got confused, and why

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT



Read the brief out once, then get out of the way. Full wording, since the slide is only the bones:

"Pick one actuator. Encode three messages: arrived, something wrong, finished. Vary only the rhythm of the pulses and how strong they are. Then hand it to another group with the screen turned away and see if they can name all three."

If someone asks to add a second actuator, say no. The constraint is the exercise.

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,
you built a visual interface with a motor glued to it.



The line worth repeating twice.

Run the test: swap groups, receiver looks away, guess all three.

Then have each group name which two got confused and what would have separated them. That confusion is the actual learning, not the successful ones.

TEST IT

Hand your device to another group with the screen turned away.

- They can tell all three messages apart.
- It does not fire continuously when held.
- You know which two got confused, and what would have separated them.
- It still works when they are not looking at it.



Put this up while the blind tests are running so groups can check themselves without asking you.

The second line matters most: a device that fires continuously when held is the single most common bug, and it is a debounce problem.

WHAT YOU ACTUALLY LEARN

Three things you cannot get from a datasheet.

What things feel like

- A motor is a nudge. A solenoid is a knock. A Peltier is slow.

How to drive one

- PWM, direction, debouncing, and why a motor never goes straight to a pin.

What to pick

- Choosing the actuator before designing the interaction, not in week 46.



PROTOTYPING WORKSHOP

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The closing argument. Thirty seconds.

If they leave with only one thing, it should be the third: pick the actuator before designing the interaction. Discovering in week 46 that the chosen one cannot do the job is the expensive mistake this session exists to prevent.



Point them at the repo. Every task sketch is commented for them, and the bench test sketches are there too if they want to see the minimal version.

Last thing to say: pick your actuator before you design the interaction, not after. Today was so they can do that from experience rather than a datasheet.

Pack down: everything back in the box, including the resistors. It will not all come back.

If anyone wants to borrow parts for the project weeks, now is when they ask